

Day 36 - Regression in Machine Learning

1) What is Simple Linear Regression?

Simple Linear Regression is a supervised machine learning algorithm used to find the relationship between one independent variable (X) and one dependent variable (Y).

It is used to predict continuous values.

Example:

Predicting salary based on years of experience.

Equation:

$$Y = mX + c$$

Where:

Y = dependent variable

X = independent variable

m = slope

c = intercept

2) What is Multiple Linear Regression?

Multiple Linear Regression is used when there are two or more independent variables to predict one dependent variable.

Example:

Predicting house price based on:

Area

Number of rooms

Location score

Equation:

$$Y = b_0 + b_1 X_1 + b_2 X_2 + b_3 X_3 + \dots + b_n X_n$$

Where:

Y = output

X_1, X_2, X_3 = input variables

b_0 = intercept

b_1, b_2, b_3 = coefficients

3) What is the mathematical implementation of Linear Regression? Show with one example
Linear Regression tries to fit the best straight line through the data points.

Formula:

$$Y = mX + c$$

Example: YouTube Ads vs Views

Suppose:

Ad Spend (X)	views (Y)
--------------	-----------

1	2
---	---

2	4
---	---

3 6

4 8

From this, the relationship is:

$$Y=2x$$

So if Ad Spend = 5, then:

$$Y=2(5)=10$$

Prediction:

If YouTube ad spend is 5 units, predicted views are 10 units.

Mathematical Goal:

Linear regression tries to find the best values of m and c such that prediction error is minimum.

4) What is the mathematical implementation for Multiple Linear Regression? Show with example

Multiple Linear Regression uses multiple inputs to predict one output.

Formula:

$$Y=b_0+b_1 x_1+b_2 x_2$$

Example: House Price Prediction

Suppose:

x_1 = Area in sq.ft

x_2 = Number of rooms

Assume equation is:

$$Y = 50000 + 300x_1 + 10000x_2$$

If:

Area = 1000 sq.ft

Rooms = 2

Then:

$$Y = 50000 + 300(1000) + 10000(2)$$

$$Y = 50000 + 300000 + 20000$$

$$Y = 370000$$

Prediction:

Predicted house price = ₹3,70,000

5) What is Loss Function?

A Loss Function is used to measure how much error the model is making.

It tells us the difference between:

Actual value

Predicted value

Purpose:

To reduce the error and improve the model.

Example Loss Functions:

Mean Squared Error (MSE)

Mean Absolute Error (MAE)

In simple words:

Loss function shows how wrong the model is.

6) What is Polynomial Regression?

Polynomial Regression is a type of regression used when the relationship between input and output is not a straight line.

It fits a curved line instead of a straight line.

Equation:

$$Y = a + bX + cX^2 + dX^3 + \dots$$

Example:

Student study hours and marks may not always follow a straight line.

Sales growth over time may be curved.

Use:

When data is nonlinear.

7) What is Optimization?

Optimization is the process of finding the best values of model parameters so that the error is minimum.

In regression:

Optimization helps find the best line or curve.

Goal:

- Minimize Loss function

- Improve prediction accuracy

Example:

Finding the best values of m and c in Linear regression.

8) What are the Evaluation Metrics for Regression Algorithms?

Evaluation metrics are used to measure how well a regression model performs.

Common metrics are:

- Mean Squared Error (MSE)

- Mean Absolute Error (MAE)

- Root Mean Squared Error (RMSE)

- R^2 Score (Coefficient of Determination)

These metrics compare:

- Actual values

- Predicted values

9) What is Gradient Descent?

Gradient Descent is an optimization algorithm used to minimize the loss function.

It helps the model learn the best values of parameters step by step.

How it works:

- Start with random values

- calculate error

- Update values slowly

- Repeat until error becomes small

In simple words:

Gradient Descent helps the model move toward the lowest error.

10) What are the Assumptions of Linear Regression?

Linear Regression works well only if some assumptions are satisfied.

Assumptions are:

- Linear relationship between input and output

- No multicollinearity between independent variables

Homoscedasticity (constant variance of errors)

Normal distribution of errors

Independence of observations

No strong outliers

These assumptions help the model perform correctly.

11) What is Overfitting Problem?

Overfitting happens when the model learns the training data too well, including noise and unnecessary details.

Result:

Very good performance on training data

Poor performance on testing data

In simple words:

The model memorizes instead of learning.

12) What is Underfitting Problem?

Underfitting happens when the model is too simple and cannot learn the data properly.

Result:

Poor performance on training data

Poor performance on testing data

In simple words:

The model fails to understand the pattern.

13) How to take the degree in Polynomial Regression?

The degree in polynomial regression decides how much the curve bends.

Examples:

Degree 1 $\rightarrow$ Straight line

Degree 2 $\rightarrow$ Quadratic curve

Degree 3 $\rightarrow$ cubic curve

How to choose degree?

Start with small degree like 2 or 3

check model performance

Avoid very high degree because it may cause overfitting

In simple words:

Choose degree based on data pattern and accuracy.

14) What is Mean Squared Error (MSE)?

MSE is the average of the squared differences between actual and predicted values.

Formula:

$$MSE = 1/n \sum (y_i - \hat{y}_i)^2$$

Where:

y_i = actual value

$\hat{y}_i$ = predicted value

Use:

It gives more importance to large errors.

15) What is Mean Absolute Error (MAE)?

MAE is the average of the absolute differences between actual and predicted values.

Formula:

$$MAE = 1/n \sum |y_i - \hat{y}_i|$$

Use:

It tells the average prediction error.

16) What is Root Mean Squared Error (RMSE)?

RMSE is the square root of Mean Squared Error.

Formula:

$$RMSE = \sqrt{MSE}$$

Use:

It shows prediction error in the same unit as output.

Example:

If house price is predicted in rupees, RMSE is also in rupees.

17) What is Lasso Regression? Why to use this?

Lasso Regression is a regularization technique that adds L1 penalty to the model.

Purpose:

- Reduce overfitting

- Perform feature selection

- Remove less important features

Why use it?

When we have many features and want to keep only important ones.

In simple words:

Lasso helps simplify the model by removing unnecessary features.

18) What is Ridge Regression? When to use this?

Ridge Regression is a regularization technique

that adds L2 penalty to the model.

Purpose:

- Reduce overfitting

- Shrink coefficient values

- Improve model stability

When to use it?

When:

- There are many features

- Features are highly correlated

- We want to reduce model complexity

In simple words:

Ridge keeps all features but reduces their impact.

19) What is the Main Goal in Regression Problems?

The main goal of regression is to predict continuous numerical values as accurately as possible.

Examples:

- Predict house price

- Predict salary

- Predict sales

- Predict temperature

Main goal:

To find the relationship between input and output and make accurate predictions.

20) What is Supervised Learning?

Supervised Learning is a type of machine learning in which the model learns from labeled data.

This means:

Input is given

correct output is also given

Examples:

classification

Regression